4. Find the vertex, focus, equation of directrix and length of the latus rectum of the following:
   (i) $y^2 = 16x$ (ii) $x^2 = 24y$ (iii) $y^2 = -8x$
   (iv) $x^2 - 2x + 8y + 17 = 0$ (v) $y^2 - 4y - 8x + 12 = 0$

5. Identify the type of conic and find centre, foci, vertices, and directrices of each of the following:
   (i) $\frac{x^2}{25} + \frac{y^2}{9} = 1$ (ii) $\frac{x^2}{3} + \frac{y^2}{10} = 1$ (iii) $\frac{x^2}{25} - \frac{y^2}{144} = 1$ (iv) $\frac{y^2}{16} - \frac{x^2}{9} = 1$

6. Prove that the length of the latus rectum of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{2b^2}{a}$.

7. Show that the absolute value of difference of the focal distances of any point P on the hyperbola is the length of its transverse axis.

8. Identify the type of conic and find centre, foci, vertices, and directrices of each of the following :
   (i) $\frac{(x-3)^2}{225} + \frac{(y-4)^2}{289} = 1$ (ii) $\frac{(x+1)^2}{100} + \frac{(y-2)^2}{64} = 1$ (iii) $\frac{(x+3)^2}{225} - \frac{(y-4)^2}{64} = 1$
   (iv) $\frac{(y-2)^2}{25} - \frac{(x+1)^2}{16} = 1$ (v) $18x^2 + 12y^2 - 144x + 48y + 120 = 0$
   (vi) $9x^2 - y^2 - 36x - 6y + 18 = 0$

5.4 Conic Sections

In addition to the method to determine the curves discussed in Section 5.3, geometric description of a conic section is given here. The graph of a circle, an ellipse, a parabola, or a hyperbola can be obtained by the intersection of a plane and a double napped cone. Hence, these figures are referred to as conic sections or simply conics.

5.4.1 Geometric description of conic section

A plane perpendicular to the axis of the cone (plane C) intersecting any one nape of the double napped cone yields a circle (Fig. 5.40) . The plane E, tilted so that it is not perpendicular to the axis, intersecting any one nape of the double napped cone yields an ellipse (Fig. 5.40). When the plane is parallel to a side of one napes of the double napped cone, the plane intersecting the cone yields a parabola (Fig. 5.41). When the plane is parallel to the plane containing the axis of the double cone, intersecting the double cone yields a hyperbola (Fig. 5.42).

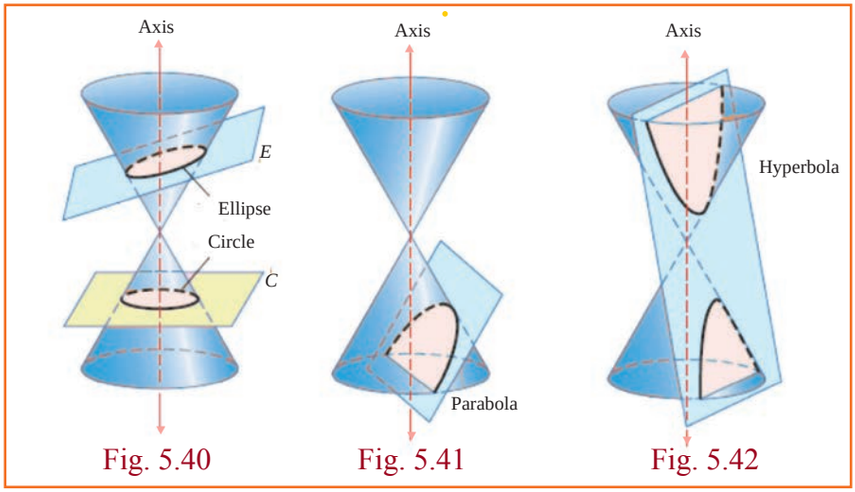


Fig. 5.40 Fig. 5.41 Fig. 5.42